

LAKEVILLE AIRLAKE, MN, USA

WMO: 726562

Lat: 44.628N

Lon: 93.228W

Elev: 960

StdP: 14.19

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04943

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.9	-8.7	-20.4	1.9	-11.7	-16.5	2.3	-8.5	25.9	14.4	23.3	16.7	5.5	280	0.532

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	90.0	74.6	86.5	72.6	83.7	71.4	78.0	85.6	75.6	82.9	73.7	80.7	9.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.3	137.4	82.6	73.0	127.0	79.9	71.9	122.2	79.0	42.1	86.9	39.8	82.9	37.9	80.7	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.3	19.1	17.0	-19.9	94.8	7.3	3.3	-25.1	97.2	-29.4	99.1	-33.5	100.9	-38.7	103.3	
			DB												
			WB	-20.2	81.4	7.1	3.1	-25.3	83.6	-29.4	85.4	-33.4	87.1	-38.5	89.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.2	14.8	18.7	31.3	45.2	57.7	68.0	71.7	68.7	61.6	48.3	34.1	20.7	
	DBStd	22.26	13.21	13.17	12.50	10.15	8.72	6.61	5.70	5.68	8.35	9.35	10.67	12.33	
	HDD50	4337	1092	876	590	205	26	0	0	0	9	146	486	907	
	HDD65	7817	1557	1296	1044	596	260	42	11	28	160	525	926	1372	
	CDD50	2586	0	0	11	62	264	539	671	579	357	94	9	0	
	CDD65	591	0	0	0	3	33	131	217	142	58	7	0	0	
	CDH74	5400	0	0	2	47	398	1184	1994	1154	543	77	1	0	
	CDH80	1626	0	0	0	11	126	374	648	302	148	17	0	0	
Wind	WSAvg	6.8	7.6	7.8	7.8	8.6	7.7	6.2	5.0	4.5	5.5	6.8	7.5	7.2	
Precipitation	PrecAvg	28.9	0.8	0.7	1.8	2.5	3.5	4.1	3.7	3.7	3.1	2.2	1.7	0.9	
	PrecMax	41.1	2.8	2.5	4.1	6.2	6.2	9.4	11.6	6.9	8.1	5.3	5.3	4.4	
	PrecMin	14.1	0.1	0.0	0.3	0.3	0.7	0.3	1.0	0.7	0.3	0.2	0.0	0.1	
	PrecStd	6.2	0.7	0.6	1.0	1.3	1.3	2.3	2.1	1.7	1.9	1.5	1.3	0.8	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.8	52.1	70.0	80.8	90.1	91.5	93.2	90.6	89.8	82.0	67.5	51.8	
		MCWB	38.8	46.7	59.1	60.6	68.5	73.6	79.3	75.7	72.1	65.1	54.9	46.6	
	2%	DB	37.5	44.9	61.2	72.4	82.5	88.3	90.1	86.4	84.0	74.2	60.9	44.9	
		MCWB	34.2	40.5	53.6	55.3	64.5	71.9	77.6	74.2	70.7	61.9	51.8	40.9	
	5%	DB	35.8	39.3	54.7	66.4	78.9	84.0	86.4	82.4	80.9	69.6	54.7	39.2	
		MCWB	33.6	36.3	49.1	53.0	62.3	69.5	74.7	71.9	68.0	58.5	48.1	36.0	
	10%	DB	32.4	36.6	48.0	62.9	72.9	81.2	83.5	80.9	76.1	63.8	48.4	36.5	
		MCWB	31.2	34.4	43.7	50.7	59.3	67.0	73.2	70.6	65.8	55.4	43.7	34.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.2	46.4	61.0	62.2	72.6	77.9	82.7	79.7	75.8	69.6	58.9	48.2
			MCDB	43.8	49.6	66.4	75.9	84.6	88.0	89.9	86.0	84.1	76.9	65.9	50.7
2%		WB	35.5	40.7	55.2	58.6	68.2	74.8	79.4	76.7	73.0	64.2	53.4	40.5	
		MCDB	37.6	44.1	59.7	68.8	78.2	83.5	87.1	83.3	80.0	70.1	59.0	44.1	
5%		WB	33.4	37.3	49.5	55.0	64.9	72.1	76.6	74.5	70.6	60.0	48.4	37.0	
		MCDB	35.3	39.4	55.0	63.7	73.1	81.1	84.1	80.7	77.5	67.1	54.2	39.3	
10%		WB	31.0	34.4	43.7	51.9	62.3	69.7	74.5	72.4	67.7	56.1	44.9	34.6	
		MCDB	32.4	36.7	48.1	60.3	70.3	77.5	81.4	78.3	74.9	63.2	48.4	36.4	
Mean Daily Temperature Range		5% DB	MDBR	16.0	16.9	16.9	19.9	20.4	19.1	19.6	19.8	21.6	19.3	15.9	14.2
			MCDBR	16.8	17.3	24.1	29.0	27.8	23.4	22.0	22.0	25.4	27.9	24.1	17.2
	MCWBR		14.3	14.0	16.4	15.6	13.0	10.9	11.7	12.0	13.4	15.6	16.2	14.2	
	5% WB	MCDBR	15.8	15.8	22.7	24.8	22.5	20.3	20.0	19.6	21.5	22.7	22.5	16.1	
		MCWBR	13.8	13.0	16.3	15.4	12.9	11.7	12.0	12.3	13.1	14.8	15.9	13.5	
Clear-Sky Solar Irradiance	taub	0.270	0.288	0.321	0.376	0.404	0.403	0.404	0.404	0.377	0.323	0.293	0.269		
	taud	2.420	2.351	2.383	2.290	2.267	2.313	2.323	2.322	2.363	2.508	2.502	2.439		
	Ebn at Noon	273	286	290	281	275	275	273	269	267	271	263	261		
	Edh at Noon	23	30	33	39	42	40	39	38	34	25	21	20		
All-Sky Solar Radiation	RadAvg	550	896	1184	1449	1688	1886	2004	1674	1312	840	537	389		
	RadStd	66	94	114	154	138	153	76	118	95	111	52	39		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page